0, 1, 2, 3 ⇒ Basics

Link

3 ⇒ qn

4

Lec_4
Practice Questions on Time Complexity

4 => Lec-3

Time complexity

{How many times Loop is executed or we Entered inside to loop}

5.) Main()

```
{
  while(n>=1)
  {
     n=n-2
  }
}
```

n = 12

12 >= 1

10 >= 1

8 >= 1

6 >= 1

4 >= 1

2 >= 1

0 > 1

n = 10

n = 8

n = 6

n = 4

n = 2

n = 0

n = 12 - 2

n = 10

n/2

$n \Rightarrow \underline{12}, \underline{10}, \underline{8}, \underline{6}, \underline{4}, \underline{2}, \underline{0}$

$n - 2k \approx 0$

$\underline{n - 2k = 1}$

$\Rightarrow n = 2k$

$\boxed{k = n/2}$

$n/2$

$O(n/2)$

$\Rightarrow O(n)$

$n = n - 2$

$n/2$

$\Rightarrow n - 3$

$\Rightarrow$

$n/3$

$n - 100$

$\Rightarrow n/100$

$O(n/100)$

$O(n/3)$

$O(n)$

$n = n - 10$

$n/10$

6.) Main()

```
{
  while(n>=1)
   {
       n=n-5
    }
}
```

20, 15, 10, 5, 0

$20/5 = 4$

$O(n/5) \Rightarrow O(n)$

$n/5$

$n - 5(k) = 0$

$k = n/5$

$n - a$ $\Rightarrow$

$a = \text{constant}$

while $(n \geq 1)$

$\{$

$n = n - a$

$\}$ $\Rightarrow O(n/a)$

$n+1 \Rightarrow O(n)$

$n+20 \Rightarrow O(n)$

$n/1000 \Rightarrow O(n) = O(n)$

$O(1), O(\log n)$

7.) Main()

```
{
  while(n>=1)
  {
    n=n-200
    n=n-100
    n=n+50
  }
}
```

1000

$n \geq 1$

$n = \cancel{1000}$

$\cancel{800}$

$\cancel{700}$

750

$n = n - 200$

cur

pre

$n = n - 300$

$n = n - 300 + 50$

$n = n - 250$

$n = 750$

1, 2, 3

$n = n - 250$

$n - a$

n/a

$O(n)$

$O(n/250)$

$n = n - 100$

$n = n - 200$

$n = n + 1000$

$n = n + 50$

$n = n + 700$

8.) Main()

```
{   i=1
    while(i<=n)
    {
        i=i+3
        i=i+5
        i=i+9
    }
}
```

$i = i + 17$

$i = 1 \quad 4 \quad 9 \quad 16 \Rightarrow 17$

$i = 10 \quad 21 \quad 26 \quad 35$

$i = 36 \quad 45 \quad 52$

$O(n/17) = O(n)$

$n/17$

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 20

$n/2$

$O(n/2) = O(n)$

Ans

```
9.) Main()
  {   i=1
    while(i<=n^2)
    {
        i=i+3
        i=i+5
        i=i+9
    }
  }
```

n^4 $i=1$,

$i<=n^2$

$i = i + 17$

$\frac{n}{17}$

$\frac{n^2}{}$

$\frac{n^4}{17}$

$\Rightarrow O(n^2)$

$\to O(n^4)$